THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 5: **Life Processes** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In photosynthesis, the oxygen released comes from: 1

(a) glucose **(b)** chlorophyll **(c)** water **(d)** carbon dioxide

2. Which has the longest small intestine? 1

(a) birds **(b)** carnivores like tiger
(c) herbivores such as the cow **(d)** humans

3. The blood vessel carrying deoxygenated blood from the heart to the lungs is: 1

(a) vena cava **(b)** aorta **(c)** pulmonary vein **(d)** pulmonary artery

4. Which of the following is NOT a requirement for photosynthesis? 1

(a) water **(b)** oxygen **(c)** chlorophyll **(d)** sunlight

5. Anaerobic respiration releases energy in the: 1

(a) ribosome **(b)** mitochondria **(c)** chloroplast **(d)** cytoplasm

6. Normal blood pressure (systolic/diastolic) in human beings is: 1

(a) 100/100 mm Hg **(b)** 120/80 mm Hg **(c)** 140/90 mm Hg **(d)** 80/120 mm Hg

7. Stomata are open during the day because: 1

(a) guard cells take up water and swell **(b)** light is absent
(c) leaves fall **(d)** guard cells lose water

8. Compared with anaerobic respiration, aerobic respiration releases: 1

(a) much less energy per molecule of glucose **(b)** the same energy
(c) no energy **(d)** much more energy per molecule of glucose

9. Which of the following transports water and minerals upward? 1

(a) cortex **(b)** epidermis **(c)** phloem **(d)** xylem

10. A leaf is kept in the dark for 24 hours and then tested with iodine. It will show: 1

(a) no blue-black colour **(b)** green colour
(c) red colour **(d)** blue-black colour

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does a leaf kept in dark not show starch? How is this tested? 2

12. Why do herbivores have a longer small intestine than carnivores? 2

13. What is the role of the lymph in the body? 2

14. Explain why the rate of photosynthesis increases with light intensity up to a limit. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the leaf is adapted for photosynthesis. 3

16. Explain why the oxygenated and deoxygenated blood must not mix in the heart. 3

17. What happens in the kidney if a person has a kidney failure? Mention treatment. 3

OR

Why is the stomach wall not digested by its own acid? 3

18. Explain the function of valves in the heart and veins. 3

OR

Explain why diffusion is insufficient for gas transport in large animals. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe an experiment to show that sunlight is necessary for photosynthesis or that CO_2 is needed, with the observations. 5

OR

Compare the human digestive system of herbivores and carnivores. 5

20. Explain digestion in the human mouth, stomach and small intestine with the enzymes and end products. 5

OR

What are the differences between aerobic respiration and anaerobic respiration? Give the steps in the cytoplasm and mitochondria. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND